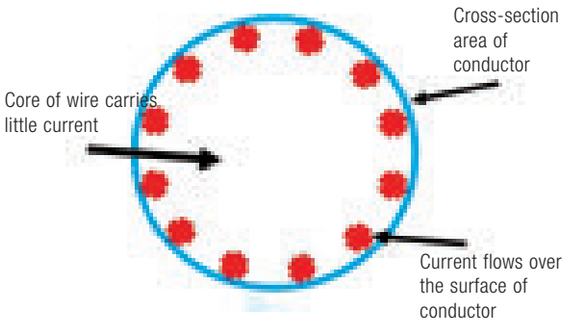


Fig. 1.8 Skin effect

surface. The high reactance of center area causes the alternating current to flow near the surface of the conductor. This is called skin effect. Due to skin effect, the effective area of cross-section of the conductor through which current flow is reduced. When the diameter of the wire is increased, skin effect will also increase.

1.7.2 Corona



Fig. 1.9 Corona

When an alternating voltage is applied across two conductors, whose space is large in comparison with the diameter, the atmospheric air is subjected to electrostatic stresses. So, a faint luminous bluish glow appears to the conductors. This bluish discharge is known as corona.

a. Factors affecting corona

i. Atmosphere
ii. Size of conductor
iii. Spacing between conductors
iv. Line voltage

b. Advantages of corona

i. Due to corona formation, the air surrounding the conductor becomes conducted and hence virtual diameter of the conductor is increased. Due to this, the static voltage between conductors is reduced.
ii. Corona reduces the effect of transient produced by surges.

c. Disadvantages of corona

i. The corona is accompanied by power loss. This affects the transmission efficiency of the line.
ii. Ozone is produced by corona. The conductor may corrode due to chemical action of ozone.
iii. The inductive current interferes with the nearby telecommunication transmitters.
iv. Most of the areas where the dirty and rag in the conductor, the light is produced.
v. When corona occurs, charging current increases due to the harmonic current.

1.8 Types of overhead lines

The transmission line has three parameters, resistance, inductance and capacitance distributed uniformly along the whole length of the line. The resistance and inductance forms series impedance. The capacitance existing between conductor for single phase line and a conductor, to

BEE_ENG_Chapter_01.indd 7 12/27/2019 10:17:26 AM